NeuroFoot

Smart Plantar Pressure Monitoring for Diabetic Neuropathy

BME 555a — USC Viterbi School of Engineering — Spring 2026

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Clinical Need & Use Cases

Section 1.1a — Stakeholder Use-Case Scenarios

Unaware Patient

A patient with loss of protective sensation walks for hours with dangerous pressure concentrated under the metatarsal heads. Without feedback, tissue damage develops silently.

Real-Time Alert

The NeuroFoot detects sustained overload and sends a WARN alert to the patient's smartphone. The patient offloads pressure before injury occurs.

Clinical Review

During a follow-up visit, the clinician reviews weeks of pressure trend data, identifies high-risk zones, and adjusts the patient's care plan accordingly.

Stakeholder Analysis & Priority

Section 1.1a — Who the device serves and why

SH-01	**Neuropathic Patient** *Direct risk of harm — highest priority per ISO 14971*	Primary alert recipient; daily wear and charging
SH-02	**Caregiver** *ADA identifies caregivers as essential in high-risk populations*	Secondary alert recipient; assists with offloading
SH-03	**Clinician** *IWGDF 2023 recommends periodic pressure monitoring*	Reviews longitudinal data; modifies care plan
SH-04	**FDA CDRH** *Defines constraints: 21 CFR 890.5575, Class II*	Regulatory reviewer for premarket submission

MVP Features Priority Ranking

Section 1.1a — Features derived from stakeholder needs

#	Feature	Priority
1	**16-Zone Plantar Pressure Sensing** FSR array across toes, metatarsal heads, midfoot, heel — core clinical function	Critical
2	**Real-Time Threshold Alerting** WARN >600 ADC, CRITICAL >850 ADC — behavioral offloading prompt	Critical
3	**BLE Wireless to iOS App** nRF52840 BLE 5.0 to companion app — <1s alert latency	Critical
4	**6-Axis IMU Gait Analysis** BMI270 accel/gyro at 50 Hz — step count, cadence, stance timing	High
5	**Ambient Environmental Monitoring** HS300x temp (±0.25°C) and humidity (±3.5% RH) — skin maceration context	Medium
6	**Data Logging for Clinician Review** Longitudinal pressure trends stored for clinical follow-up	Medium

MVP Specifications Linked to Features

Section 1.1b — Measurable performance requirements

Feature	Specification	Verification Method
16-zone pressure sensing	≥16 zones, 50 Hz, 0–400 kPa, $R^2 > 0.90$	Calibrated weight test, MuxPressureTest MODE 1
Threshold alerting	WARN >600 ADC within 1s, CRITICAL >850 within 1s, clear <2s	Controlled load, measure response time
BLE communication	Advertise <3s, connect <10s, 4 characteristics at rate	Power-on to app data display <15s
IMU gait analysis	±1g (±0.05g) accel, <±0.5 dps gyro bias	6-position static cal, zero-rate bias test
Environmental sensing	Temp ±0.5°C vs reference, humidity ±5% RH	Side-by-side with calibrated instruments

Regulatory: Intended Use & Predicate Device

Section 1.1c — 21 CFR 890.5575, Product Code IRN, Class II

SmartStep (K060150)

Predicate Device — Andante Medical Devices

- Flexible pressure-sensing insole
- Belt-worn Control Unit + PC Software
- Threshold-based overload alarm
- Rehabilitation / partial weight-bearing
- Prescription Use under physician supervision
- Cleared Feb 23, 2006 (Class II)

NeuroFoot (Proposed)

Team Cardinal — USC BME 555a

- 16-zone FSR array in sandal form factor
- Arduino Nano 33 BLE + iOS companion app
- Graded alerts: OK / WARN / CRITICAL
- Continuous daily ambulatory monitoring
- Prescription Use under clinician supervision
- 510(k) pathway via K060150 predicate

Substantial Equivalence & Clinical Alignment

Section 1.1c — Key SE comparisons and guideline reconciliation

Feature	SmartStep	NeuroFoot	Equivalence
Sensor	Resistive insole	16-zone FSR via mux	Same modality
Pressure	Single threshold	16-zone 50 Hz + PTI	Enhanced resolution
Alerts	Audible/visual alarm	OK/WARN/CRITICAL + haptic	Same core function
Wireless	Control Unit to PC	BLE 5.0 to iOS	Both wireless
Form Factor	Insole + belt unit	Sandal + embedded MCU	Both wearable
Safety Std	IEC 60601-1, ISO 14971	Same + IEC 62304	Same core standards

Clinical Guideline Alignment

IWGDF 2023: Pressure offloading for neuropathic patients • ADA 2024: Comprehensive foot exams + early intervention • APMA: Wearable pressure-offloading technologies

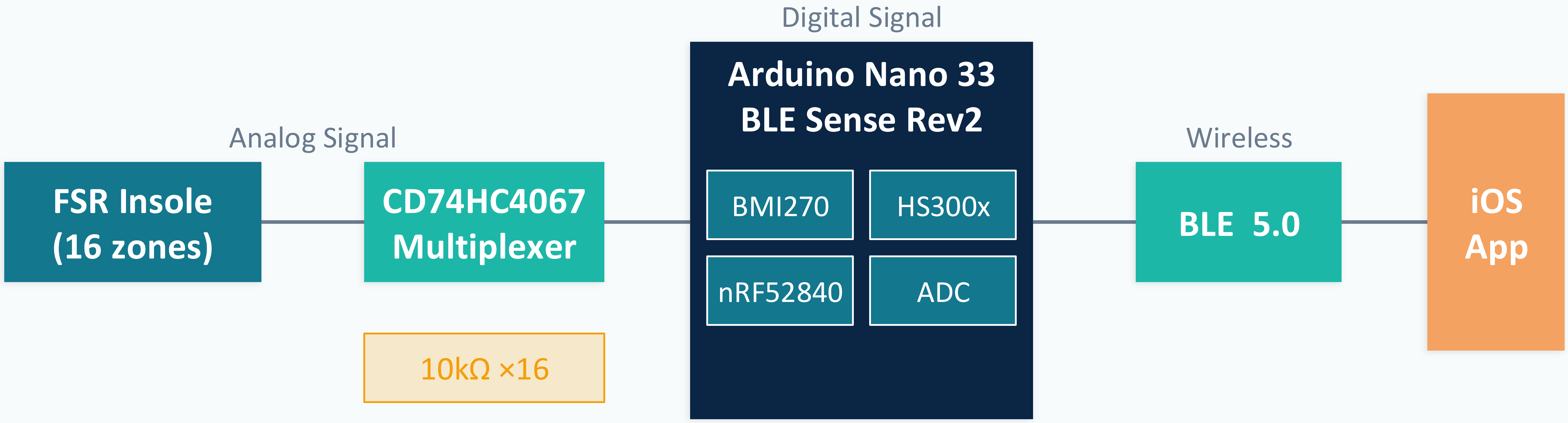
Bill of Materials Overview

Section 1.2a — Key components and specifications

Component	Part Number / Description	Key Specification	Qty
Microcontroller	Arduino Nano 33 BLE Sense Rev2 (ABX00070)	nRF52840, BLE 5.0, BMI270, HS300x	1
Multiplexer	CD74HC4067 (16-ch analog)	Break-before-make, 100µs settle	1
FSR Insole	In-house developed 16-zone sensor	Resistance varies 10kΩ	1
Pull-down Resistors	10kΩ	Voltage divider for FSR channels	16
Sandal	EVA foam, two-layer construction	Board housed between layers	1 pair

Hardware Block Diagram

Section 1.2a — System architecture and signal flow



Software Decision Logic & App Workflow

Section 1.2a — Firmware thresholds and iOS companion app

Firmware Decision Logic

Read 16 FSR channels (50 Hz scan)

Check each zone against thresholds

< 600: OK	600–849: WARN	≥ 850: CRIT

Record duration each zone spends within thresholds

Transmit via BLE to iOS app

iOS Companion App

- Real-time 16-zone pressure heatmap
- Zone bar charts with threshold markers
- Duration each zone spends within a threshold recorded
- Gait metrics: steps, cadence, stance
- Ambient temperature & humidity display
- WARN/CRITICAL alert banners + haptic
- Connection status & battery monitoring
- SwiftUI + CoreBluetooth framework

Acceptance Criteria

Section 1.2b — Per-component and integrated system verification

Component	Test	Pass Criteria	Result
Arduino Nano 33 BLE	Power-on self-test	Board ID + BLE MAC within 5s	✓ Pass
CD74HC4067 Mux	Channel isolation	<5 ADC unloaded, no cross-talk	✓ Pass
FSR Array (16 zones)	Response linearity	Monotonic, $R^2 > 0.90$	✓ To Be Tested
BMI270 Accel	6-position static cal	±1g (±0.05g)	✓ Pass
BMI270 Gyro	Zero-rate bias	<±0.5 dps stationary 30s	✓ Pass
HS300x Temp	Reference comparison	±0.5°C vs calibrated	✓ Pass
BLE Link	Connection test	Connect <10s	✓ Pass
Alert System	Threshold trigger	WARN <1s, CRITICAL <1s	✓ Pass
Full System	2-min walk test	Continuous data, no dropouts	✓ To Be Tested/Improved
BMM150 Magneto	I2C read	Valid axis readings	✗ **Not Included**

Test Plan & Calibration Methods

Section 1.2b — Verification sequence and calibration approach

Integration Verification

Stage 1 — Breadboard

Tested Arduino - IMU (accel and gyro), Temp, and Humidty

Arduino + MUX + FSR on breadboard. MuxPressureTest sketch confirms 16 independent channels with 10kΩ pull-downs. NeuroFoot_BLE sketch verifies iOS app receives all 4 BLE characteristics.

Stage 2 — Sandal Assembly

Transfer to sandal form factor. Arduino secured between EVA layers. FPC breakout connected via ribbon cable. Full walk test verifies no signal degradation, loose connections, or BLE range issues.

Calibration Methods

- FSR: Calibrated weights (0, 5, 10, 20, 40 kg) on rigid surface, ADC vs load curve
- IMU Accel: 6-position static test, 100 samples per orientation, mean per axis
- IMU Gyro: Stationary 30s, 3000 samples at 100 Hz, compute zero-rate bias
- Temp: Side-by-side with NIST-traceable thermometer, 5-min stabilization
- Humidity: Side-by-side with calibrated hygrometer, 5-min stabilization
- BLE: Power-on to app data display timing, repeated 10× for consistency

Traceability & Components to Avoid

Section 1.2c — Breadboard → Altium → PCB, and lessons learned

Design Requirement — BOM Component — Breadboard Validation — Altium Schematic — PCB Vendor

🚫 Components & Configurations to AVOID

- 100kΩ pull-downs — compressed ADC range to ~270 counts; use 10kΩ instead
- Piezoelectric sensors — no sustained reading under static load; unsuitable for plantar monitoring
- BMM150 magnetometer — not accessible on Rev2 Wire1 bus; excluded from design
- Wire.begin() in firmware — corrupts onboard I2C bus (HS300x, BMI270 fail)
- APDS9960 init before other I2C — causes bus conflicts; must initialize last

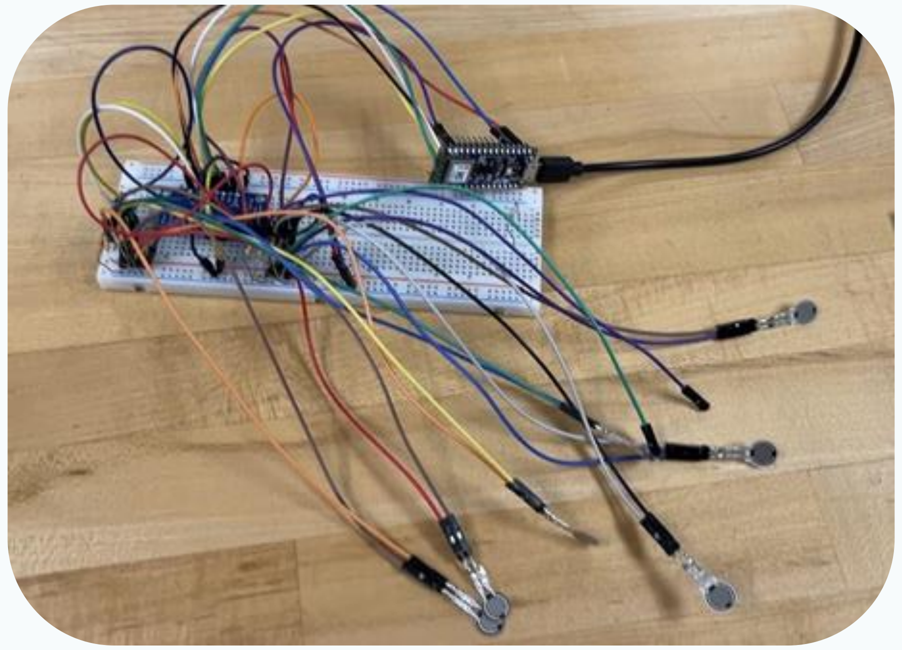
Manufacturing Process Flow

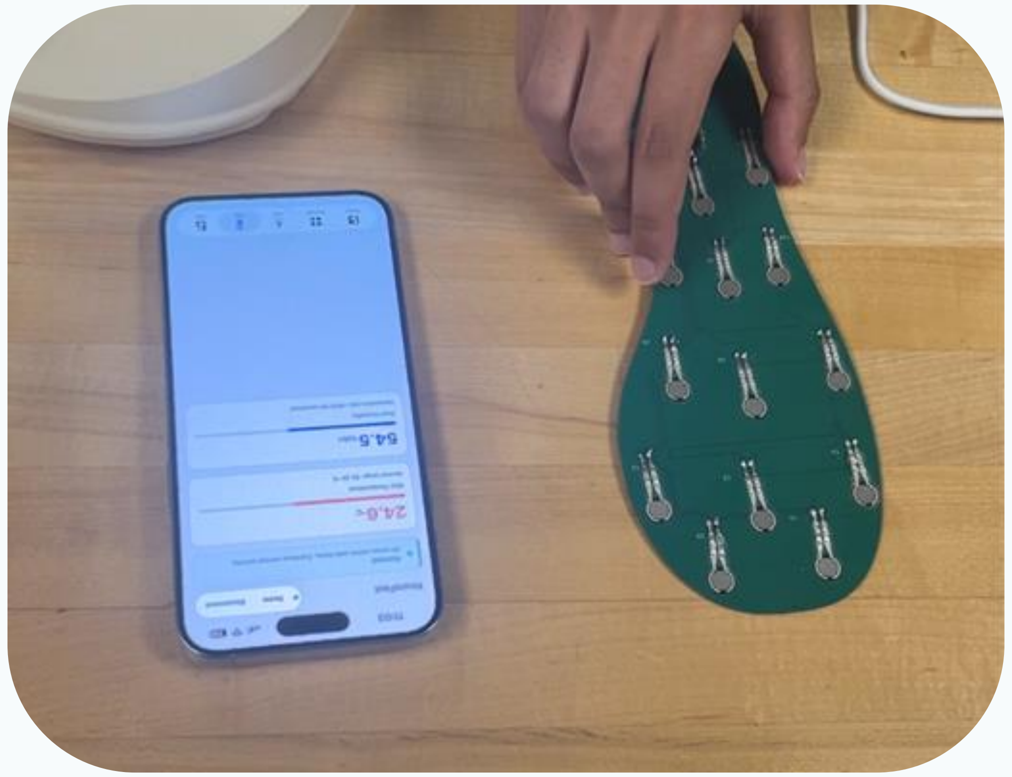
Section 1.2d — Repeatable assembly recipe

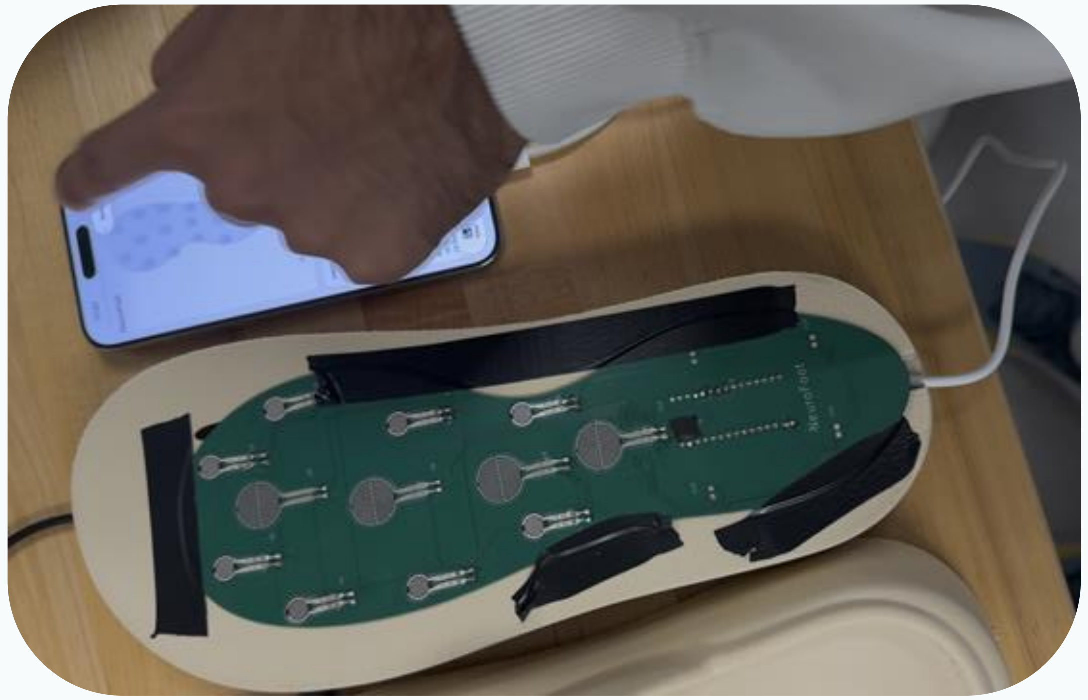
1. Populate 16 × 10kΩ pull-downs on breadboard/PCB
2. Mount CD74HC4067 mux; wire S0–S3 to Arduino digital pins
3. Connect FPC breakout board; attach insole ribbon cable
4. Wire mux SIG → Arduino A0; VCC/GND connections
5. Flash firmware (NeuroFoot_BLE sketch); verify serial output
6. Run MuxPressureTest — validate all 16 channels
7. Cut sandal in half; seat Arduino + PCB between layers; glue
8. Route FPC cable; verify no pinch points; close sandal
9. Full system walk test — confirm no signal degradation

Prototype Verification & Testing

Section 1.2e — Assembly images and verification results







Verification Summary

9/10 components passed acceptance testing. BMM150 magnetometer rejected (I2C inaccessible on Rev2). Integration verified in two stages: breadboard validation followed by sandal assembly walk test. All pressure channels, IMU axes, environmental sensors, BLE link, and alert thresholds confirmed operational.

Identified Failure Modes

Section 2.1 — Six failure modes across five categories

Failure Mode	Category	Clinical Impact	Source
FSR adhesive delamination	Mechanical Wear	Underreported pressure → missed alerts	Anticipated
60 Hz electromagnetic interference	Environmental	False threshold crossings → alert fatigue	**Observed**
Moisture ingress / corrosion	Environmental	Short circuits, signal degradation	Anticipated
BLE disconnection	Wireless	No alerts reach patient during overload	Anticipated
User forgets to charge	User Error	Device dies mid-use, all monitoring stops	Anticipated
HS300x cold-boot zero read	Firmware	Erroneous 0.0 values, reduced confidence	**Observed**

2 of 6 failure modes were directly observed and resolved during prototype development. The remaining 4 are anticipated based on the device's mechanical and electrical design.

FMEA Identified Failure Modes

Section 2.1 — Six failure modes across five categories

FM-01 · Mechanical

FSR adhesive delamination

Underreported pressure; missed alerts during sustained overload

Source: Anticipated

FM-02 · Environmental

60 Hz EMI on analog lines

False alerts from noise; patient ignores legitimate warnings

Source: Anticipated

FM-03 · Environmental

Moisture ingress / corrosion

Short circuits; silent monitoring loss during wear

Source: Anticipated

FM-04 · Wireless

BLE disconnection

No alerts reach patient during dangerous loading

Source: Observed

FM-05 · User Error

User forgets to charge

Complete monitoring loss for unknown duration

Source: Anticipated

FM-06 · Firmware

HS300x cold-boot zero read

Erroneous 0.0 values on first read; no alert impact

Source: Observed

Risk Analysis — Severity Scale (S)

Section 2.2a — Worst-case consequence if failure reaches patient undetected

Score	Level	Clinical Context
1–3	Minor	Function slightly degraded; patient self-corrects; no safety impact
4–6	Moderate	Accuracy reduced or alert frequency affected; no direct tissue injury
7–8	Serious	Core monitoring fails; patient exposed to sustained overload without warning
9–10	Critical	Direct physical injury or complete silent failure; patient believes device works

Scoring Rationale

Scores reflect consequences for the NeuroFoot intended use population: adults with diabetic peripheral neuropathy and loss of protective sensation (LOPS). A score of 7+ means the patient cannot independently detect the failure.

Risk Analysis — Occurrence Scale (O)

Section 2.2a — Estimated probability of failure during expected device lifetime

Score	Frequency	Basis
1–2	Remote (<1 in 10,000)	Component spec provides very high reliability assurance
3–4	Low (~1 in 1,000)	Component meets spec; failure possible but infrequent
5–6	Moderate (~1 in 100)	Known failure mechanism in literature; design partially mitigates
7–8	High (~1 in 10)	Failure mechanism dominant; current design does not address
9–10	Very High (>1 in 2)	Failure expected under normal use without corrective action

Estimates based on available evidence at PDR stage: datasheet specs, published literature, and prototype test observations.

Risk Analysis — Detection Scale (D)

Section 2.2a — Ability of design controls to identify failure before patient impact

Score	Detectability	Mechanism
1–2	Almost Certain	Firmware/hardware auto-detects; visible alert before patient impact
3–4	High	Patient would likely notice anomalous behavior within one session
5–6	Moderate	Requires active inspection or app review; not self-evident during use
7–8	Low	Gradual or latent failure; patient unlikely to notice without monitoring
9–10	Remote	Silent failure; device gives no indication of malfunction

Higher score = lower probability of detection. A score of 10 means the device gives no indication of failure — the patient cannot distinguish operational from non-operational.

Risk Acceptance Criteria

Section 2.2a — RPN-based classification and required actions

200–1000 HIGH	**Corrective action required before Design Verification**	Revised RPN must fall below 200 after controls implemented
100–199 MEDIUM	**Risk reduction recommended**	Actions documented and addressed in Design Verification plan
1–99 LOW	**Acceptable with existing controls**	Monitor and reassess if design changes are made

Risk Analysis — RPN Scoring

Section 2.2 — Severity x Occurrence x Detection

S (Severity) **O (Occurrence)** **D (Detection)** **RPN = S x O x D**

ID	Failure Mode	S	O	D	RPN	Risk
FM-01	FSR adhesive delamination	9	5	7	**315**	**HIGH**
FM-02	60 Hz EMI false alerts	5	4	6	**120**	**MEDIUM**
FM-03	Moisture ingress / short circuit	7	5	7	**245**	**HIGH**
FM-04	BLE disconnection	7	4	6	**168**	**MEDIUM**
FM-05	Battery depletion / user failure	7	5	5	**175**	**MEDIUM**
FM-06	HS300x cold-boot init failure	2	4	3	**24**	**LOW**

FDA & ISO Risk Management Alignment

ISO 14971: RPN >=200 (FM-01, FM-03) require mandatory design controls. **ISO 13485 7.1:** All risk controls documented and verified. **FDA 21 CFR 820.30(g):** Design validation must address identified risk scenarios.

Risk Mitigation & Design Controls

Section 2.3 — Risk reduction strategies and residual RPN

FM-01 · Adhesive Delamination

RPN: 315 -> 54

Medical-grade adhesive + recessed midsole channel + firmware self-test

FM-02 · 60 Hz EMI

RPN: 120 -> 20

10k pull-downs (confirmed) + PCB ground plane + 3-sample debounce

FM-03 · Moisture Ingress

RPN: 245 -> 42

Conformal coating + IP44 enclosure + sealed FPC connector

FM-04 · BLE Disconnection

RPN: 168 -> 24

Onboard buzzer for CRITICAL + auto-reconnect every 5s + visible LED

FM-05 · Battery Depletion

RPN: 175 -> 42

Audible low-battery alarm + caregiver push notification + wireless charging

FM-06 · HS300x Init Failure

RPN: 24 -> 8

Firmware retry loop (3x, 500ms delay) — already deployed in prototype

Mitigations & Design Controls

Section 2.3 — Risk reduction strategies per failure mode

Failure Mode	Risk Control Measure	Type
Adhesive delamination	Snap-fit insole mount; medical-grade adhesive; peel-strength test	Design
60 Hz EMI	10k pull-downs; 100nF bypass caps; shielded PCB traces	Design
Moisture ingress	Conformal coating; sealed FPC connector; hydrophobic liner	Design
BLE disconnection	Onboard buzzer + LED; app auto-reconnect; BLE loss alert	Firmware
User charging failure	Audible low-battery alarm; app notification; USB-C quick charge	Design + UX
HS300x cold-boot	Firmware retry loop (3x, 100ms delay); validated in prototype	Firmware

Summary & Fall 2026 Handoff

Team Cardinal · NeuroFoot System · April 27, 2026

WHAT WAS BUILT & VERIFIED

✓ 16-zone FSR insole + CD74HC4067 mux — all channels verified
✓ BLE 5.0: pressure (50 Hz), IMU (50 Hz), environmental (0.5 Hz)

✓ Graded alerts (OK/WARN/CRITICAL) — <200 ms latency

✓ HS300x temp (+-0.5C) and humidity (+-5% RH) verified

✓ IMU 6-position accel cal and gyro zero-rate bias confirmed

✓ Sandal assembly — 2-min walk test, no BLE dropouts

✓ Predicate: SmartStep (K060150) — SE documented

✓ ISO 13485-compliant DFM documentation

FALL 2026 — NEXT TEAM CHECKLIST

IMMEDIATE

→ Implement FMEA mitigations

PCB DESIGN

→ Translate to Altium (see traceability)

→ No 100k pull-downs. No Wire.begin(). No BMM150.

VALIDATION

→ Dynamic force cal with reference plate

→ IEC 60601-1-2 EMC testing

→ Human walking trials (5-30 subjects)

REGULATORY

→ File 510(k) with K060150 predicate

Thank You
Questions?